

Complex Dynamics and High codimension Bifurcations of a Predator Prey Model

Wael El Khateeb

University of Toledo

In collaboration with Dr. Chunhua Shan, Dr. Guihong Fan, and Dr. Hao Wang.



History



Adriatic Sea: East of Italy

During World War I, fishing in regions such as the Adriatic Sea declined dramatically, leading to unusual fluctuations in fish populations. Vito Volterra became intrigued by this phenomenon.

Fluctuation in population numbers

Fishermen noticed that the populations of certain prey fish and their predators did not simply increase or decrease together; instead, they appeared to oscillate in a regular, cyclical fashion.



The Bee-Mite Problem



Bee with Varroa mites.

Florida Insects

<http://bit.ly/4hpAqiz>

Real-Life Data



Classical Model

The Lotka–Volterra model is a pair of first-order, nonlinear differential equations that describe the dynamics of predator and prey populations. Let $x(t)$ represent the prey population and $y(t)$ represent the predator population. The model is given by:

$$\begin{cases} \frac{dx}{dt} = \alpha x - \beta xy, \\ \frac{dy}{dt} = \delta xy - \gamma y, \end{cases} \quad (1)$$

where:

- α is the natural growth rate of the prey,
- β is the predation rate coefficient,
- δ is the growth rate of predators per prey consumed, and
- γ is the natural death rate of the predators.

Extension of Classical Model

Instead of exponential growth, logistic growth is used to account for resource limitations:

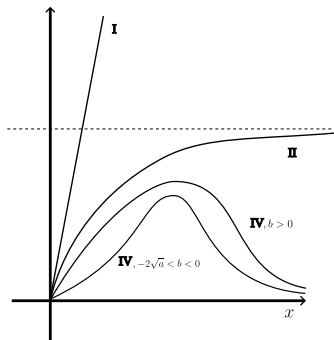
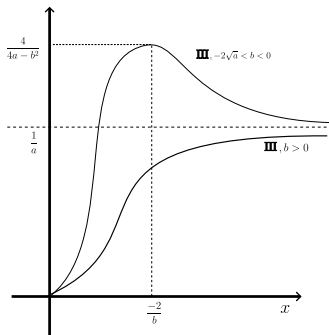
$$\frac{dx}{dt} = \alpha x \left(1 - \frac{x}{K}\right) - p(x)y,$$

where K is the carrying capacity. We can incorporate the natural harvesting of prey by subtracting h :

$$\frac{dx}{dt} = \alpha x(t) \left(1 - \frac{x(t)}{K}\right) - p(x)y - h.$$

Holling Functional Response

Describe how a predator's rate of prey consumption changes with prey density. They were developed by Canadian ecologist C. S. Holling in the 1950s and 1960s.



Linearization-ODE 1

Definition (Fixed Point)

A point (x^*, y^*) is called a *fixed point* of the dynamical system

$$\begin{cases} \dot{x} = f(x, y) \\ \dot{y} = g(x, y) \end{cases}$$

if it satisfies $f(x^*, y^*) = g(x^*, y^*) = 0$

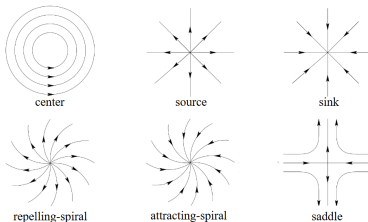
Theorem (Hartman-Grobman)

Near a hyperbolic fixed point E , the dynamics of a nonlinear system are topologically conjugate to its linearization.

Linearization ODE 2

$$\begin{cases} \frac{dx}{dt} = f(x, y), \\ \frac{dy}{dt} = g(x, y), \end{cases} \rightarrow J = \begin{bmatrix} f_x(x^*, y^*) & f_y(x^*, y^*) \\ g_x(x^*, y^*) & g_y(x^*, y^*) \end{bmatrix} \rightarrow \lambda^2 - \text{Tr}(J)\lambda + \det(J) = 0$$

Based on the eigenvalues, we can deduce the type of equilibria.

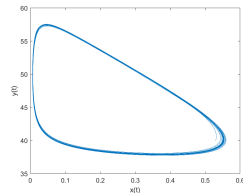
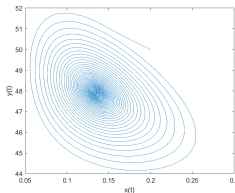
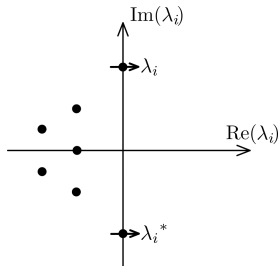


VERMA V., KAO D., PANG A.: A flow-guided streamline seeding strategy. In Proceedings of the conference on Visualization'00 (2000), IEEE Computer Society Press, pp. 163–170

Local Hopf Bifurcation

Definition

A *Hopf bifurcation* occurs when a fixed point of a dynamical system loses stability as a pair of complex conjugate eigenvalues crosses the imaginary axis, resulting in the emergence or disappearance of a periodic orbit of small amplitude.



Translation of the predation action

Predators need time to reproduce after predation, which is why we add a delay.

$$\begin{cases} \frac{dx}{dt} = rx(1 - \frac{x}{K}) - p(x(t - \tau))y, \\ \frac{dy}{dt} = p(x(t))y - dy. \end{cases} \quad (2)$$

τ represents the time required for the predator's consumption of prey to translate into an increase in the predator population. The mortality of predators during this period is accounted for by the function $g(\tau, d)$:

$$\begin{cases} \frac{dx}{dt} = rx(1 - \frac{x}{K}) - p(x(t))y, \\ \frac{dy}{dt} = g(\tau, d)p(x(t - \tau))y - dy. \end{cases} \quad (3)$$

Equilibria

- The system exhibits three equilibria: $E_0 = (0,0)$, $E_K = (K,0)$, and E^* .
- E^* is an interior point.
- All orbits with initial conditions in first quadrant stay in the first quadrant.

ODE vs. DDE

Ordinary Differential Equations (ODEs):

- The rate of change depends only on the current state.
- Typically formulated as $\frac{dx}{dt} = f(x, t)$. (Suitable for systems with instantaneous interactions)

Delay Differential Equations (DDEs):

- The rate of change depends on past states as well as the current state.
- Typically formulated as $\frac{dx}{dt} = f(x(t), x(t - \tau), t)$ with delay $\tau > 0$. The initial conditions for DDEs are defined in an interval of time $(-\tau, 0)$.

Functional Differential Equation

Define

$$C = C([-r, 0], \mathbb{R}^n)$$

Let

$$\sigma \in \mathbb{R}, r, A \in \mathbb{R}^+, x \in C([\sigma - r, \sigma + A], \mathbb{R}^n), \text{ and } t \in [\sigma, \sigma + A]$$

We define $x_t(\theta) = x(t + \theta)$, $\theta \in [-r, 0]$. Assume Ω is a subset of $\mathbb{R} \times C$, $f: \Omega \rightarrow \mathbb{R}^n$ is a given function, then we call:

$$\dot{x}(t) = f(t, x_t) \tag{4}$$

a retarded delay differential equation or functional differential equation. Initial data for the delay differential equations are generally continuous functions defined on a finite interval.

Existence and Uniqueness of Solutions

Theorem (Existence)

Suppose that Ω is an open subset in $\mathbb{R} \times C$ and f is continuous on Ω . If $(\sigma, \phi) \in \Omega$, then there is a solution of (1.1) passing through (σ, ϕ) .

Theorem (Uniqueness)

Suppose Ω is an open set in $\mathbb{R} \times C$, $f: \Omega \rightarrow \mathbb{R}^n$ is continuous and $f(t, \phi)$ is Lipschitz in ϕ in each compact set in Ω . If $(\sigma, \phi) \in \Omega$, then there is a unique solution of Equation 4 through (σ, ϕ)

Note: We say $f(t, \phi)$ is Lipschitz in ϕ in a compact set K of $\mathbb{R} \times C$ if there is a constant $k > 0$ such that for any $(t, \phi_i) \in K, i = 1, 2$,

$$|f(t, \phi_1) - f(t, \phi_2)| \leq k|\phi_1 - \phi_2|$$

Linearization DDE 1

$$\begin{cases} \frac{dx}{dt} = f(x, y, x(t-\tau), y(t-\tau)) \\ \frac{dy}{dt} = g(x, y, x(t-\tau), y(t-\tau)) \end{cases} \rightarrow \lambda^2 + C_1\lambda + e^{-\lambda\tau}(C_2\lambda + C_3) + C_4 = 0$$

Possibly infinitely many eigenvalues satisfy the characteristics equation. In our case, we get

$$P(\lambda, \tau) := \lambda^2 + [d - H(\tau)]\lambda + e^{-\lambda\tau}[-d\lambda + L(\tau)] - dH(\tau) = 0, \quad (5)$$

where $H(\tau)$ and $L(\tau)$ are continuous functions of τ .

Multiplicity of roots

Lemma

The sum of the multiplicity of roots of $P(\lambda, \tau) = 0$ on $\mathbb{C}^+ = \{\lambda \in \mathbb{C} : \operatorname{Re}(\lambda) > 0\}$ can change only if a pair of conjugate complex roots appear on or cross the imaginary axis as τ varies continuously on $[0, \tau_{\max})$.

Finding Bifurcation points

We are interested in finding the delay thresholds that get us pure imaginary eigenvalues, ie, $\lambda = i\omega$. We replace λ by $i\omega$ in the characteristics equation:

$$-\omega^2 + [d - H]i\omega + e^{-i\tau\omega}[-di\omega + L] - dH = 0.$$

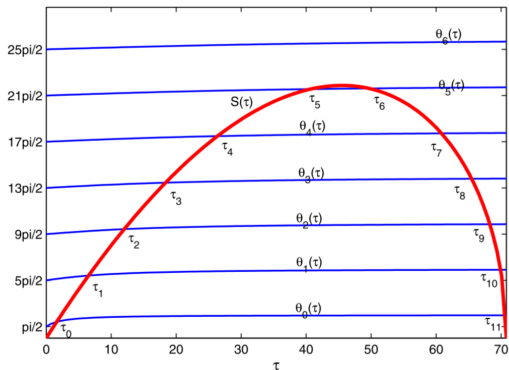
Using Euler's Formula and the trigonometric identity, we get:

$$\begin{cases} \sin(\tau\omega) = \frac{d\omega L - H\omega L - d^2\omega H - d\omega^3}{L^2 + d^2\omega^2}, \\ \cos(\tau\omega) = \frac{LdH + L\omega^2 + (d\omega)^2 - dH\omega^2}{L^2 + d^2\omega^2}. \end{cases} \quad (6)$$

$$\omega(\tau) = \sqrt{\frac{\sqrt{((H(\tau))^4 - 4d^2(H(\tau))^2 + 4(L(\tau))^2)} - \frac{(H(\tau))^2}{2}}{2}}. \quad (7)$$

Finding Bifurcation points 2

Accordingly, we are interested in finding the points (ω, τ) that satisfy both equations!



Condition for Hopf Bifurcation

Theorem (Beretta and Kuang(2002))

$$\operatorname{sign}\left\{\left.\frac{d}{d\tau}\operatorname{Re}(\lambda)\right|_{\lambda=i\omega(\tau^*)}\right\}=\operatorname{sign}\left\{\left.\left(\theta'-(\omega(\tau)\tau)'\right)\right|_{\tau=\tau^*}\right\}$$

Analysis of the graph at hand

To properly analyze the graph above, we should check our eigenvalues at $\tau = 0$.

We therefore have three possibilities that we can conclude from the graph at hand. Denote by $\tau = c_0$ the value over which $S(\tau)$ attains its maximum.

- if $S(c_0) > \theta_0(c_0)$, the graph has $2k$ intersections between $w\tau$ and θ . Thus we expect $2k$ hopf bifurcation thresholds since the difference $(S' - \theta')$ is different than zero, and hence the transversality condition is satisfied.

Remaining Cases

- if $S(c_0) = \theta_0(c_0)$, then there exists one pure imaginary eigenvalue that does not cross to the right side of the plane. Therefore, we do not have Hopf bifurcation.
- If $S(c_0) < \theta_0(c_0)$, then the two graphs of S and θ do not intersect. this means that all eigenvalues would stay on the left side of the plane. $N=N^*$ would be globally asymptotically stable.

Global Hopf Bifurcation

Define the closed set $\Sigma(F)$ in $C([-1, 0], \mathbb{R}_+^2) \times (0, \tau_{max}) \times \mathbb{R}_+$ as follows:

$$\Sigma(F) = Cl\{(z, \tau, T) \in C([-1, 0], \mathbb{R}_+^2) \times (0, \tau_{max}) \times \mathbb{R}_+ \mid z$$

is a T -periodic solution of the DDE system. Let $C(E^*, \tau_n^{(i)}, \frac{2\pi}{\theta(\tau_n^{(i)})})$ be the connected component of $(E^*, \tau_n^{(i)}, \frac{2\pi}{\theta(\tau_n^{(i)})})$ in $\Sigma(F)$.

Global Hopf Bifurcation Theory

Theorem

Consider the original Functional Differential Equations system, and suppose they satisfy a series of Conditions C1-4. Then one of the following two statements holds.

(i) $C(E^, \tau_n^{(i)}, \frac{2\pi}{\theta(\tau_n^{(i)})})$ is unbounded;*

(ii) $C(E^, \tau_n^{(i)}, \frac{2\pi}{\theta(\tau_n^{(i)})})$ is bounded, the intersection*

$C(E^, \tau_n^{(i)}, \frac{2\pi}{\theta(\tau_n^{(i)})}) \cap N(F)$ is finite, and*

$$\sum_{(z, \tau, T) \in C(E^*, \tau_n^{(i)}, \frac{2\pi}{\theta(\tau_n^{(i)})}) \cap N(F)} \gamma_1(z, \tau, T) = 0.$$

Here $i = 1, 2, \dots, \chi(n)$ and $n \in \mathbb{N}$.

Bounded Period

Lemma

If $K < K_0$, the DDE system has no periodic solutions of period 1.

Lemma

If $K < K_0$, then $T_n^{(i)} \in (\frac{1}{n+1}, \frac{1}{n})$ for $i = 1, 2, \dots, \chi(n)$ and $n = 1, 2, \dots$.

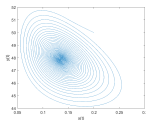
Nested Structure

Theorem

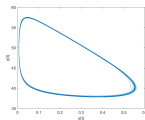
Under the assumptions taken earlier, along with $K < K_0$, the following two statements hold.

- (1) All connected components $C(E^*, \tau_n^{(i)}, \frac{2\pi}{\theta(\tau_n^{(i)})})$ are bounded for all $n = 1, 2, \dots$.*
- (2) If $\chi(n) = 2$, denote $\tau_n^{(1)}$ and $\tau_n^{(2)}$ as τ_n^- and τ_n^+ , respectively. Then the connected components $C(E^*, \tau_n^-, \frac{2\pi}{\theta(\tau_n^-)})$ and $C(E^*, \tau_n^+, \frac{2\pi}{\theta(\tau_n^+)})$ coincide and are nested. The connected component $C(E^*, \tau_n^\pm, \frac{2\pi}{\theta(\tau_n^\pm)})$ connects exactly two bifurcation points τ_n^- and τ_n^+ , where $n = 1, 2, \dots$.*

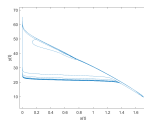
Global Hopf Bifurcation Simulations



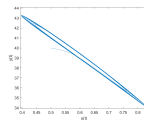
$\tau = 0.5$



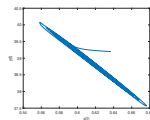
$\tau = 0.7$



$\tau = 8.5$

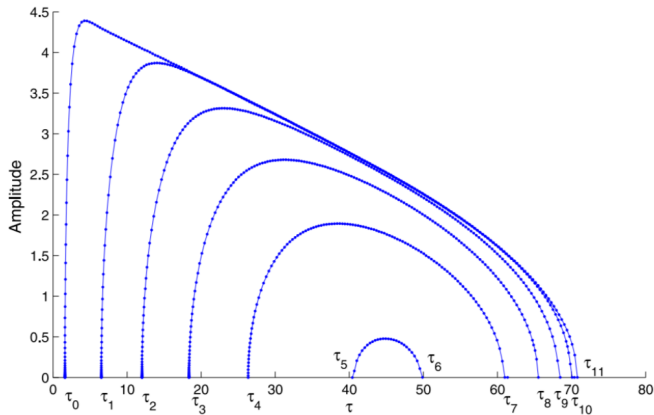


$\tau = 14.4$

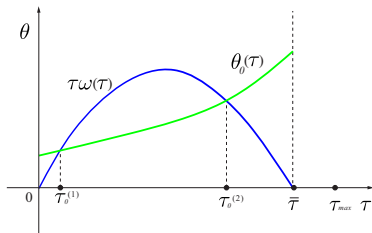


$\tau = 14.8$

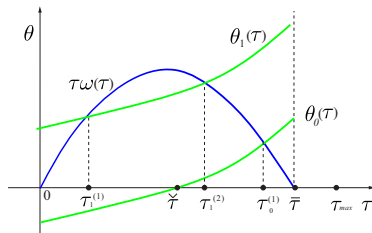
Amplitude vs delay



Novel Results 1

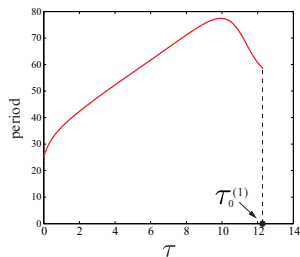
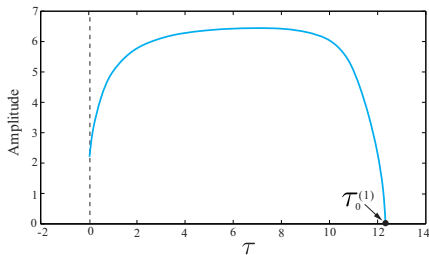


$$K < K_0$$

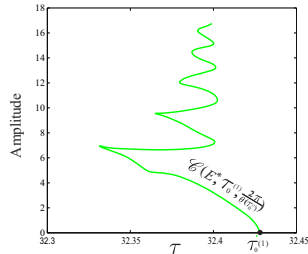
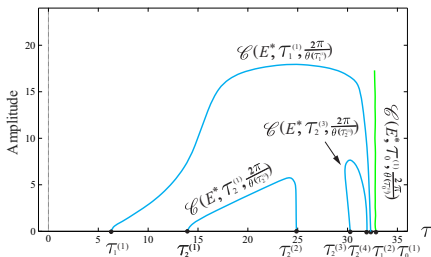


$$K > K_0$$

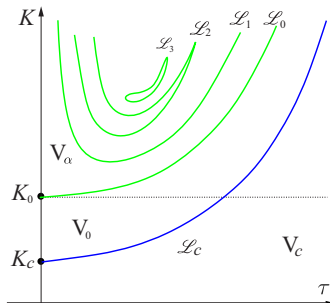
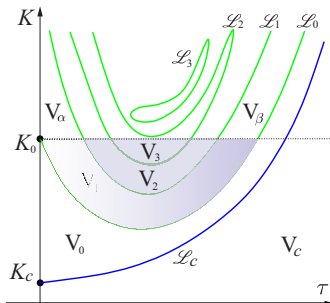
Continuity of Branches



Novel Results 2



Novel Results 3



Summary

References

- Shu, H., Wang, L., Wu, J. (2013). Global dynamics of Nicholson s blowflies equation revisited: Onset and termination of nonlinear oscillations. *Journal of Differential Equations*, 255(9), 2565-2586.
- Kuang, Y. (Ed.). (1993). *Delay differential equations: with applications in population dynamics*. Academic press.
- Fan, G., Wolkowicz, G. S. (2021). Chaotic dynamics in a simple predator-prey model with discrete delay.
- Wu, J. (1998). Symmetric functional differential equations and neural networks with memory. *Transactions of the American mathematical society*, 350(12), 4799-4838.
- Liu, B., Teng, Z., Chen, L. (2006). Analysis of a predator-prey model with Holling II functional response concerning impulsive control strategy. *Journal of Computational and Applied Mathematics*, 193(1), 347-362.
- Holling, C. S. (1965). The functional response of predators to prey density and its role in mimicry and population regulation. *The Memoirs of the Entomological Society of Canada*, 97(S45), 5-60.